

Barcoding Detection and Parameter Sensitivity

OCT Profile Extraction Proof-of-Concept

Jonathan Ma

July 31, 2026

Roadmap

- 1 Interactive Profile Overlay and Manual Classification
- 2 Profile Extraction Tests
- 3 Normalization Scopes
- 4 One-Factor-at-a-Time Analysis
- 5 Factorial Analysis
- 6 Conclusions

New Tool: Interactive Profile Overlay

Purpose

The annotation interface aligns the processed sub-BM scan with its extracted intensity profile on a shared horizontal axis.

Workflow

- 1 Select an annotation class:
 - Normal
 - Early Atrophy (EA)
 - Barcoding
- 2 Drag horizontally across either:
 - the processed OCT scan, or
 - the aligned intensity profile.
- 3 Use `Undo last` or `Clear all` to revise intervals.
- 4 Save the annotations when enabled.

Saved information

- Source E2E file
- Central B-scan index
- Profile depth and step size
- Profile margin and aggregation method
- Normalization configuration
- Interval start and end positions
- Class label and interval width

Manual Classification Results

Initial proof-of-concept annotation

Three contiguous horizontal regions were manually classified on the central B-scan.

Class	Start x	End x	Width (px)
Normal	2.0	131.5	129.5
Early Atrophy (EA)	137.1	185.0	47.9
Barcoding	191.9	263.5	71.6

Interval counts

- Normal: 1
- Early Atrophy: 1
- Barcoding: 1

Total annotated width

- Normal: 129.5 px
- Early Atrophy: 47.9 px
- Barcoding: 71.6 px

Profile Extraction Expansion

We want to evaluate whether averaging neighboring rows produces a more robust intensity profile than extracting a single horizontal line.

Configurations evaluated

Method	Description
Single Line	Baseline profile from one row
Mean ± 2 px	Average across 5 rows
Mean ± 4 px	Average across 9 rows
Median ± 2 px	Median across 5 rows
Median ± 4 px	Median across 9 rows
Median ± 6 px	Median across 13 rows

Summary statistics

- Gradient energy — amount of rapid intensity change, profile roughness.
- Correlation — similarity to the original single-line extraction (1.0 = identical).
- (MAD) — average point-by-point intensity difference from the baseline.
- (RMSD) — overall magnitude of deviation, emphasizing larger differences.

Profile Extraction Results

Method	Gradient	Correlation	MAD	RMSD
Single Line	270.8	1.000	0.00	0.00
Mean ± 2	108.0	0.933	7.45	9.27
Mean ± 4	81.4	0.908	8.60	10.76
Median ± 2	146.3	0.935	6.15	9.11
Median ± 4	107.2	0.907	7.82	10.76
Median ± 6	83.5	0.901	8.35	11.11

Observations

- Increasing the averaging band substantially reduced roughness.
- Mean and median aggregation produced similar overall profiles.
- Median ± 2 produced the smallest deviation from the baseline profile.
- Larger averaging windows generated smoother profiles but differed more.

Clinical interpretation

- Averaging neighboring rows suppresses local speckle noise while preserving the overall profile shape.
- Small averaging windows (± 2 pixels) appear to provide the best balance between noise reduction and preservation of local features.
- Very large averaging windows may smooth clinically relevant local intensity changes associated with early atrophy or barcoding.

Current recommendation

For subsequent experiments, a profile band of approximately ± 2 pixels is recommended as the baseline extraction strategy. This configuration substantially reduces profile variability while remaining highly correlated ($r \approx 0.93$) with the original single-line extraction.

Normalization Scope Comparison

Objective

Evaluate whether normalization should be performed using the entire sub-Bruch's membrane (BM) region or only the rows used for profile extraction.

Normalization strategies

Method	Description
Whole ROI	Percentiles computed using all pixels below BM
Local Band	Percentiles computed only within the extraction band

Both strategies were evaluated using:

- Robust percentile normalization (1–99%) & Min–max normalization (0–100%)

Summary statistics

- Correlation, MAD, RMSD, Gradient Energy, and dynamic range, the difference between the darkest and brightest profile values after normalization.

Normalization Scope Results

Method	Correlation	MAD	RMSD	Gradient
Whole ROI (1–99)	1.0000	0.00	0.00	180.1
Local Band (1–99)	0.9971	10.24	12.49	328.3
Whole ROI (0–100)	0.99996	39.80	41.26	84.0
Local Band (0–100)	0.99996	10.67	11.05	151.4

Normalization intensity ranges

Method	Lower	Upper
Whole ROI (1–99)	0.0	171.0
Local Band (1–99)	25.1	149.0
Whole ROI (0–100)	0.0	251.0
Local Band (0–100)	0.0	187.0

Interpretation

Clinical interpretation

- Whole-ROI normalization provides a consistent intensity reference using the entire processed scan.
- Local-band normalization adapts the intensity scaling to the immediate neighborhood surrounding the extracted profile.
- Restricting normalization to the local band increases local contrast, producing larger intensity gradients that may improve visibility of subtle features.
- Min-max normalization (0–100%) uses the full intensity range and is more influenced by extreme pixel values than percentile-based normalization.

Current recommendation

Percentile normalization using the entire processed ROI is recommended because it provides a stable global reference while remaining robust to extreme intensity values.

One-Factor-at-a-Time Experimental Setup

Objective

Measure how strongly each processing choice changes the extracted intensity profile when all other settings are held fixed.

Baseline configuration

Parameter	Baseline setting
Profile margin	± 4 pixels
Aggregation	Mean
Normalization scope	Whole ROI
Percentile bounds	1–99%
Sampling step size	1 pixel

Factors varied independently

- Profile margin: 0, ± 2 , ± 4 , ± 6 , and ± 8 pixels
- Percentile bounds: 0–100, 0.5–99.5, 1–99, 2–98, and 5–95
- Sampling step size: 1, 2, 4, and 8 pixels

How OFAT Sensitivity Was Measured

Each alternative profile was compared with the baseline profile at the same horizontal image positions.

- **Sufficient Statistics:** Correlation, MAD, RMSD, Max Absolute Difference, Grad Energy
- **Curvature energy:** the amount of rapid change in profile slope.
 - Helps measure sharp peaks and narrow local features.

OFAT Results: Profile Margin and Aggregation

Setting	Correlation	MAD	RMSD	Roughness
Margin ± 0 px	0.907	12.80	16.00	207.2
Margin ± 2 px	0.986	4.68	5.88	103.1
Margin ± 4 px	1.000	0.00	0.00	83.8
Margin ± 6 px	0.992	3.44	4.39	68.2
Margin ± 8 px	0.983	4.77	6.26	62.3
Mean aggregation	1.000	0.00	0.00	83.8
Median aggregation	0.989	4.01	5.14	99.4

Interpretation

- The single-line profile was substantially rougher and differed most from the baseline.
- Increasing the margin progressively smoothed the profile.
- Margins of ± 2 to ± 6 pixels remained highly similar to the baseline.
- Mean and median aggregation produced similar profile shapes, although the median retained slightly more local roughness.

OFAT Results: Sampling Step Size

Step size	Correlation	MAD	RMSD	Roughness
1 px	1.000	0.00	0.00	83.8
2 px	0.981	3.55	6.71	66.3
4 px	0.954	6.95	10.36	34.7
8 px	0.894	11.09	15.95	13.5

Interpretation

- Larger step sizes reduce the number of sampled horizontal positions.
- This produces progressively smoother profiles because narrow local changes may be skipped.
- A 2-pixel step remained relatively close to the baseline.
- Step sizes of 4 or 8 pixels increasingly altered the profile shape and may miss narrow barcode-like features.

Which Parameters Had the Greatest Influence?

Factor	Mean MAD	Maximum RMSD	Minimum correlation
Percentile bounds	15.80	41.26	0.998
Sampling step size	5.40	15.95	0.894
Profile margin	5.14	16.00	0.907
Normalization scope	5.12	12.49	0.997
Aggregation method	2.01	5.14	0.989

Clinical interpretation

- **Percentile bounds** changed the absolute gray-value scale most strongly, although the spatial profile shape was largely preserved.
- **Sampling step size** and **profile margin** had the greatest effect on profile shape and local detail.
- **Aggregation method** had the smallest overall influence.

Current working configuration

- Profile margin: approximately ± 4 pixels
 - Aggregation: mean
 - Normalization: whole-ROI, 1–99%
 - Sampling step size: 1 pixel
-
- Small symmetric averaging bands reduce speckle while preserving the underlying spatial profile.
 - Whole-ROI percentile normalization provides a reproducible intensity reference.
 - Native 1-pixel sampling is preferred for preserving narrow barcode structures.
 - These findings are based on the current proof-of-concept scan and require confirmation across additional OCT volumes.

Factorial Experimental Setup

Objective

Evaluate combinations of preprocessing and profile-extraction settings to determine whether parameter interactions produce substantially different intensity profiles.

Factors included

Parameter	Levels evaluated
Profile margin	± 2 , ± 4 , and ± 6 pixels
Aggregation	Mean and median
Normalization scope	Whole ROI and local band
Percentile bounds	0–100, 1–99, and 2–98%
Sampling step size	1 and 2 pixels

$3 \times 2 \times 2 \times 3 \times 2 = 72$ parameter combinations

Each resulting profile was compared with the working baseline:

± 4 pixels, mean aggregation, whole-ROI 1–99% normalization, and 1-pixel sampling

Most Stable Factorial Configurations

Margin	Aggregate	Scope	Bounds	Step	Corr.	MAD	RMSD
± 4	Mean	Whole ROI	1–99	1	1.000	0.00	0.00
± 6	Mean	Whole ROI	1–99	1	0.992	3.44	4.39
± 4	Median	Whole ROI	1–99	1	0.989	4.01	5.14
± 6	Median	Whole ROI	1–99	1	0.987	4.47	5.59
± 2	Mean	Whole ROI	1–99	1	0.986	4.68	5.88
± 4	Mean	Whole ROI	2–98	1	1.000	6.09	6.30
± 4	Mean	Whole ROI	1–99	2	0.981	3.55	6.71

Observed pattern

- The most stable configurations consistently used whole-ROI normalization.
- Percentile bounds of 1–99% appeared most frequently among the highest-ranked settings.
- Margins of ± 4 or ± 6 pixels were more stable than the smallest tested margin.
- One-pixel sampling generally produced the closest agreement with the baseline.

Configurations Producing the Largest Changes

Margin	Aggregate	Scope	Bounds	Step	Corr.	MAD	RMSD
± 6	Mean	Whole ROI	0–100	1	0.992	40.06	41.79
± 6	Mean	Whole ROI	0–100	2	0.976	39.66	41.71
± 2	Mean	Whole ROI	0–100	2	0.970	39.77	41.70
± 6	Median	Whole ROI	0–100	1	0.987	39.87	41.65
± 6	Median	Whole ROI	0–100	2	0.973	39.60	41.64
± 2	Median	Whole ROI	0–100	2	0.961	39.45	41.48

Interpretation

- Min–max normalization produced the largest gray-value shifts.
- Correlations remained relatively high, indicating that the overall spatial pattern was preserved despite substantial scale changes.
- This means two profiles can have similar shape while having very different absolute intensity values.
- Local-band 2–98% normalization also produced larger deviations, although less extreme than whole-ROI min–max normalization.

Balancing Stability and Signal Preservation

Stability alone is not sufficient: excessive smoothing may reduce noise while also suppressing clinically relevant barcode-like intensity changes.

- **Gradient energy ratio**

- A value near 1 indicates similar preservation of local intensity changes.
- Values below 1 indicate stronger smoothing.
- Values above 1 indicate greater local variation.

- **Stability–signal score**

- Lower scores indicate better agreement with the baseline while retaining a similar amount of local structure.
- This is an exploratory ranking measure rather than a clinical diagnostic score.

Configuration	RMSD	Gradient ratio	Score
± 4 , mean, whole ROI, 1–99, step 1	0.00	1.000	-2.577
± 6 , median, whole ROI, 1–99, step 1	5.59	0.974	-2.050
± 4 , mean, whole ROI, 2–98, step 1	6.30	1.094	-1.854
± 6 , mean, whole ROI, 1–99, step 1	4.39	0.814	-1.829
± 4 , median, whole ROI, 1–99, step 1	5.14	1.186	-1.766

Factorial Analysis Conclusions

Main findings

- Stable parameter combinations clustered around whole-ROI percentile normalization, small-to-moderate averaging bands, and native 1-pixel sampling.
- Percentile bounds strongly affected absolute intensity scale.
- Profile margin and sampling step size more directly affected local profile detail.
- Mean and median aggregation were both viable, with comparatively smaller effects than normalization and sampling choices.

Working configuration

Use 0.01-0.99 normalization about the whole ROI, using mean ± 4 pixels

These results describe one proof-of-concept scan and will be validated across the remaining OCT volumes before fixing final defaults.

Main findings

- Small-to-moderate depth averaging reduced profile roughness while preserving overall signal shape.
- Whole-ROI percentile normalization provided the most stable intensity reference.
- Sampling step size strongly affected preservation of narrow local features.
- Aggregation method had a smaller influence than normalization, margin, or step size.
- Stable configurations clustered around:
 - ± 4 to ± 6 pixel margins,
 - whole-ROI 1–99% normalization,
 - and 1-pixel sampling.

① Apply the pipeline to all available E2E volumes

- Test the central scan of each volume
- Test across all scans of a single volume

② Develop numerical detection features

- Local intensity level
- Gradient and curvature activity
- Peak density and spatial frequency
- Vertical persistence across neighboring depths
- Look into Neural Super Resolution (a resolution enhancing technique like DLSS)

③ Evaluate automated spatial classification

- Threshold-based signal processing
- Change-point methods
- Hidden Markov models